

Part 1: Perimeter and Area Relationships of a Rectangle

Optimization: the process of finding values that make a given quantity the greatest (or least) possible given certain conditions.

Investigation 1:

In this investigation you will determine how to find the maximum area of a rectangle with a fixed perimeter.

Remember:

$$\text{Perimeter of a Rectangle} = 2(l + w) \text{ or } 2l + 2w$$

$$\text{Area of a Rectangle} = \text{length} \times \text{width}$$

Instructions: Brandon needs to find the dimensions that will maximize the rectangular area of an enclosure with a perimeter of 32 meters.

a) Copy and complete the following table to determine the dimensions of the rectangle that maximizes the area of a rectangle that has a perimeter of 32 meters.

Rectangle	Width	Length	Perimeter	Area
1	1	15	32	15
2	2	14	32	28
3	3	13	32	39
4	4	12	32	48
5	5	11	32	55
6	6	10	32	60
7	7	9	32	63
8	8	8	32	64

b) What are the dimensions of the rectangle with the maximum area? 8×8

c) What is the maximum area? 64 m^2

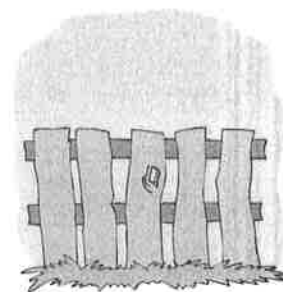
d) Describe the shape of the rectangle:

It is a square.

Problem 1

Your neighbour has asked for your advice about building his garden. He fence the largest rectangular garden possible with 12 metres of fencing. the dimensions of the garden with the maximum area?

How can you organize the possible shapes so you don't miss one that the maximum area?



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What are

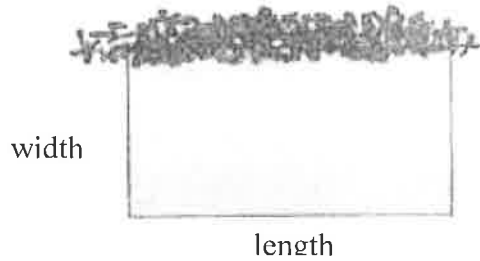
might give

Length	Width	Perimeter	Area
1	5	12	5
2	4	12	8
* 3	3	12	9 *
4	2	12	8

Answer: The dimensions that give the maximum area of 3×3 .
This gives an area of 9 m^2 .

Investigation 2: In this investigation you will determine how to find the maximum area of a rectangle with a fixed sum of the lengths of three sides.

Instructions: Brandon's customer decides to use an existing hedge as one of the boundaries for the enclosure. This means that he will only use the prefabricated fencing on three sides of the rectangular enclosure. The client still wants the enclosure to have the greatest area possible. Brandon has 32 meters of prefabricated fencing.



a) Do you think Brandon will be able to enclose more, less, or the same amount of area now that the hedge is being used on one side? *More.*

b) What shape do you think will have the maximum area?

A rectangle with the length twice the width.

c) Copy and complete the chart to determine the dimensions of the rectangle that has the maximum area.

Rectangle	Width	Length	Sum of lengths of three sides	Area
1	1	30	32	30
2	2	28	32	56
3	3	26	32	78
4	4	24	32	96
5	5	22	32	110
6	6	20	32	120
7	7	18	32	126
* 8	8	16	32	128 *
9	9	14	32	126
10	10	12	32	120

d) What are the dimensions of the rectangle with maximum/optimal area? 8x16

e) What is the maximum area? 128 m²

f) Describe the shape of the rectangle: The length is double the width.

Problem 2

A farmer is adding a rectangular corral to the side of a barn. The barn will form one side of the rectangle. The farmer has 16 meters of fencing to use. Conduct an investigation to determine the dimensions of the corral with maximum area.

$$A = (l)(w)$$

$$P = l + 2w$$

Width	Length	Perim.	Area
1	14	16	14
2	12	16	24
3	10	16	30
* 4	8	16	32 *

Width	Length	Perim	Area
5	6	16	30
6	4	16	24
7	2	16	14

The dimensions are: The dimensions that give max area are 4x8. The max area is 32 m².

Consolidation:

The dimensions of a rectangle that has maximum area depend on the number of sides to be fenced.

When fencing all four sides of a rectangle, the optimal shape is a square.

When only three sides are being fenced, the relationship between the

length and the width of the rectangle is $l = 2w$.

Apply What You Learned

1) What dimensions will provide the maximum area for a rectangle with each perimeter?

a) 20 m $20 \div 4 = 5$

A 5m x 5m square

c) 50 m

$$50 \div 4 = 12.5$$

A 12.5m x 12.5m square

b) 36 m

$$36 \div 4 = 9$$

A 9m x 9m square

d) 83 m

$$83 \div 4 = 20.75$$

A 20.75m x 20.75m square

2. Cam is building a fenced-in play yard for his small children against one wall of his house. He has 12 m of fencing. What whole-number dimensions produce the play yard with the greatest area?

using a chart:

width	length	Perimeter	Area
1	10	12	10
2	8	12	16
* 3	6	12	18*
4	4	12	16
5	2	12	10

Note: $P = l + 2w$

$$A = (l)(w)$$

using algebra:

$$P = l + 2w$$

but we know $l = 2w$

$$\text{so } P = 2w + 2w$$

we know $P = 12$

$$12 = 4w$$

$$\frac{12}{4} = w$$

$$3 = w$$

If $w = 3$ and we know area is maximized when the $l = 2w$; the length must be 6.

The dimensions that maximize the area are 3m by 6m.
The max area is 18m^2 .